

ODE

U1
§1 { 常微分方程: 单自变量
偏微分 --- : 2/2+ 自变量

微分方程中未知函数的最高阶数:
微分方程的阶

$$\text{一阶: } F(x, y, \frac{dy}{dx}) = 0 \\ \Rightarrow \frac{dy}{dx} = f(x, y)$$

$$(a, b) \quad F(x, \varphi(x), \varphi'(x)) \equiv 0$$

$y = \varphi(x)$ 为一个解.

(a, b) 定义区间

$\Phi(x, y) = 0$ 隐函数,

定理

$$\frac{dy}{dx} = f(x, y)$$

条件 $y|_{x=x_0} = y_0$ $\leftarrow$ 初值条件

$f(x, y)$ 在 $D: |x - x_0| \leq a, |y - y_0| \leq b$ 连续,

有连续偏导 $f_y'(x, y)$.

在 $|x - x_0| \leq h$

于唯一 $y = \varphi(x)$ 满足

特解

$(x_0, y_0) \in D$

其中 $h = \min(a, \frac{1}{m})$

$$M = \max_{(x,y) \in D} |f(x,y)|$$

§2

一. 可分离变量.

$\frac{dy}{dx} = \varphi(x)\phi(y)$ 可分离变量微分方程

$$\frac{dy}{\phi(y)} = \varphi(x) dx$$

积分

$$\frac{dy}{dx} = \frac{y(1-x)}{x}$$

$$\frac{dy}{y} = \frac{1-x}{x} dx$$

$$\ln|y| = \ln|x| - x + C_1$$

$$|y| = |x| e^{-x+C_1} \quad C = \pm e^{C_1}$$

$$y = Cx e^{-x}$$

二. 齐次

$$f(x,y) = g\left(\frac{y}{x}\right)$$

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right) \quad \text{零齐次微分方程.}$$

$$u = \frac{y}{x} \quad y = ux$$

$$\frac{d(ux)}{dx} = g(u)$$

$$x \frac{du}{dx} + u = g(u)$$

$$\frac{du}{dx} = \frac{g(u) - u}{x}$$

$$\frac{du}{g(u) - u} = \frac{dx}{x}$$

$$\int \frac{du}{g(u) - u} = \ln|x|$$

eg. $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$

$$u = \frac{y}{x} \quad y = ux$$

$$x \frac{du}{dx} + u = \frac{u}{1+u^2}$$

$$x \frac{du}{dx} = \frac{-u^3}{1+u^2}$$

$$-\frac{1-u^2}{u^3} du = \frac{dx}{x}$$

变量代换法.

$$\frac{1}{2u^2} - \ln|u| = \ln|ax|$$

$$\frac{x^2}{2y^2} - \ln\left|\frac{y}{x}\right| = \ln|cx|$$

$$\frac{x^2}{2y^2} = \ln|cy|$$

$$y = ce^{\frac{x^2}{2y^2}}$$

§3 — 一阶线性微分方程

$$\frac{dy}{dx} + p(x)y = f(x)$$

s.t. $f(x) \equiv 0$.

$$\frac{dy}{dx} + p(x)y = 0$$

$$e^{-\int p(x)dx} \left(\int f(x) e^{\int p(x)dx} dx + C \right)$$

$$\frac{dy}{y} = -p(x)dx$$

$$p(x)u \square$$

$$p(x)y$$

$$\ln|y| = \int p(x)dx + \ln|c|$$

$$y = u e^{-\int p(x)dx}$$

令 $u = u(x)$ 为待定函数

$$\frac{dy}{dx} = e^{-\int p(x)dx} \frac{du}{dx} - u e^{-\int p(x)dx} p(x) \quad \text{代回}$$

$$e^{-\int p(x)dx} \frac{du}{dx} - u e^{-\int p(x)dx} p(x) + p(x) u e^{-\int p(x)dx} = f(x)$$

$$\Rightarrow p e^{-\int p(x)dx} \frac{du}{dx} = f(x)$$

$$du = f(x) e^{\int p(x)dx} dx$$

$$u = \int f(x) e^{\int p(x)dx} dx + C$$

$$y = e^{-\int p(x)dx} \left[\int f(x) e^{\int p(x)dx} dx + C \right]$$

$$= \underbrace{e^{-\int p(x)dx} \int f(x) e^{\int p(x)dx} dx}_{\downarrow} + \underbrace{C e^{-\int p(x)dx}}_{\downarrow}$$

$C=0$

一个解

$$y(x) = y^*(x) + Y(x)$$

通解

$$y = e^{-\int_{x_0}^x p(\xi) d\xi} \left[\int_{x_0}^x f(\xi) e^{\int_{x_0}^{\xi} p(\zeta) d\zeta} d\xi + y_0 \right]$$

$$x \in (a, b)$$

$$x_0 \rightarrow x$$

$$\text{eg. } \frac{dy}{dx} - y \cot x = 2x \sin x$$

$$\frac{dy}{dx} - y \cot x = 0.$$

$$\ln|y| = \ln|\sin x| + \ln|c|$$

$$y = c \sin x$$

$$y = u \sin x$$

$$\frac{dy}{dx} = \sin x \frac{du}{dx} + u \cos x$$

$$\sin x \frac{du}{dx} = 2x \sin x.$$

$$\frac{du}{dx} = 2x$$

$$u = x^2 + c$$

$$y = (x^2 + c) \sin x \quad \leftarrow y|_{x=\frac{\pi}{2}} = 0.$$

$$c = -\frac{\pi^2}{4}$$

$$y = \left(x^2 - \frac{\pi^2}{4}\right) \sin x$$

$$2x \frac{dy}{dx} - y = -x$$

$$\ln$$

$$\frac{dy}{dx} - \frac{y}{2x} = -\frac{1}{2}$$

$$e^{\int \frac{1}{2x} dx} \left[-\frac{1}{2} \int e^{-\int \frac{1}{2x} dx} dx + C \right]$$

$$-\frac{1}{2} \ln|x|$$

$$\sqrt{|x|} \left[-\frac{1}{2} \int \frac{1}{\sqrt{|x|}} dx + C \right]$$

$$y = -x + C \sqrt{|x|}$$

eg. 方程 2π 周期解:

$$y(x) = y(x + 2\pi)$$

$$y(0) = y(2\pi)$$

= Bernoulli

$$\frac{dy}{dx} + p(x)y = f(x)y^n$$

$$y^{-n} \frac{dy}{dx} + p(x) y^{1-n} = f(x)$$

$$z = y^{1-n} \quad \rightarrow$$

$$\frac{dz}{dx} = (1-n) y^{-n} \frac{dy}{dx}$$

$$\frac{dz}{dx} + (1-n)p(x)z = (1-n)f(x)$$

eg. $\frac{dy}{dx} + \frac{y}{x} = a(\ln x) y^2$

$$y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = a(\ln x)$$

$$z = y^{-1}$$

$$\frac{dz}{dx} = -y^{-2} \frac{dy}{dx}$$

$$= \frac{z}{x} - a \ln x \quad -\ln x$$

$$z = e^{-\int \frac{1}{x} dx} \left[\int -a \ln x e^{\int \frac{1}{x} dx} dx + C \right]$$

$$= x \left[\int (-a \ln x) \cdot \frac{1}{x} dx + C \right]$$

$$= x \left[C - \frac{a}{2} (\ln x)^2 \right]$$

$$y = \frac{1}{x \left[C - \frac{a}{2} (\ln x)^2 \right]}$$

$$\text{eg. } \frac{dy}{dx} = \frac{y}{2x} + \frac{x^2}{2y}$$

$$\text{取 } n = -1$$

$$y \frac{dy}{dx} = \frac{y^2}{2x} + \frac{x^2}{2}$$

$$z = y^2$$

$$\begin{aligned} \frac{dz}{dx} &= 2y \frac{dy}{dx} \\ &= \frac{y^2}{x} + x^2 \end{aligned}$$

$$\frac{dz}{dx} = \frac{z}{x} + x^2$$

$$\Rightarrow y^2 = z = cx + \frac{1}{2}x^3$$

$$\equiv \text{Riccati } \frac{dy}{dx} = p(x)y^2 + q(x)y + r(x)$$

$$\text{取特解 } y = \phi_1(x)$$

→ 一般不能用初等方法求解

$$\text{令 } u(x) = y(x) - \phi_1(x)$$

$$\frac{du}{dx} = \frac{dy}{dx} - \frac{d\phi_1(x)}{dx}$$

$$= p(u + \phi_1)^2 + q(u + \phi_1) + r - p\phi_1^2 - q\phi_1 - r$$

$$= [2p\phi_1 + q]u(x) + p(x)u^2 \rightarrow \text{Bernoulli}$$

§4 全微分方程.

$$M(x, y)dx + N(x, y)dy = 0$$

$$M(x, y)dx + N(x, y)dy = du(x, y)$$

若 $y = y(x)$ 为一个解.

$$\begin{cases} M(x, y(x))dx + N(x, y(x))dy \equiv 0 \\ du(x, y(x)) \equiv 0 \\ \rightarrow u(x, y(x)) \equiv C_0 = u(x_0, y_0) \end{cases}$$

隐函数.

为全微分方程充要条件:

$$\frac{\partial M}{\partial y} \equiv \frac{\partial N}{\partial x}, (x, y) \in G.$$

法1

$$u(x, y) = \int_{(x_0, y_0)}^{(x, y)} M(x, y)dx + N(x, y)dy.$$

$$= \int_{x_0}^x M(\xi, y_0)d\xi + \int_{y_0}^y N(x, \eta)d\eta$$

法2

或 $du(x, y) = M(x, y)dx + N(x, y)dy$

$$\frac{\partial u}{\partial x} = M(x, y)$$

$$\frac{\partial v}{\partial y} = N(x, y)$$

$$u(x, y) = \int m(x, y) dx + p(y) \quad \checkmark C \quad \star$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \int m(x, y) dx + p'(y) = N(x, y)$$

$$\Rightarrow p'(y) = N(x, y) - \frac{\partial}{\partial y} \int m(x, y) dx$$

法3 配凑

$$y dx + x dy = d(xy)$$

$$\frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$$

$$\frac{-y dx + x dy}{x^2} = d\left(\frac{y}{x}\right)$$

$$\frac{-y dx + x dy}{x^2 + y^2} = d\left(\arctan \frac{y}{x}\right)$$

$$\frac{y dx - x dy}{x^2 - y^2} = d\left(\frac{1}{2} \ln \left| \frac{x-y}{x+y} \right| \right)$$

$$m(x, y) dx + N(x, y) dy = 0.$$

若 $\exists \mu(x, y) \neq 0$.

s.t. $\mu(x, y) m(x, y) dx + \mu(x, y) N(x, y) dy = 0$ 为全微分方程. ①

即 $\mu(x, y) \neq 0$.

$$\text{s.t. } \mu(x, y) M(x, y) dx + \mu(x, y) N(x, y) dy \equiv du(x, y)$$

则 $\mu(x, y)$ 为①的一个积分因子

$u(x, y) = C$ 为①通积分.

$$(y + x^2 + y^2) dx - x dy = 0.$$

$$\frac{y dx - x dy + \cancel{(x^2 + y^2) dx}}{x^2 + y^2} = 0$$

$$\text{对于 } M \frac{\partial u}{\partial y} + \mu \frac{\partial v}{\partial y} = N \frac{\partial v}{\partial x} + \mu \frac{\partial N}{\partial x}$$

若 $\mu(x, y)$ 仅与 x 或 y 有关

$$0 + \mu \frac{\partial v}{\partial y} = N \frac{\partial v}{\partial x} + \mu \frac{\partial N}{\partial x}$$

$$\mu \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = N \frac{d\mu}{dx}$$

$$\frac{1}{\mu} \cdot \frac{d\mu}{dx} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$$

$$\text{同理取 } \mu(y) \quad \frac{1}{\mu} \cdot \frac{d\mu}{dy} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M}$$

$$\text{若 } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \text{ 仅与 } x \text{ 有关} = h(x)$$

$$\text{积分因子取 } \mu(x) = e^{\int h(x) dx}$$

$$\text{若 } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} \text{ 仅与 } y \text{ 有关} = h(y)$$

$$\text{积分因子取 } \mu(y) = e^{\int h(y) dy}$$

$$\text{eg. } \underbrace{(-y + xy^2)}_M dx + \underbrace{xy}_{N} dy = 0$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -2(1 - xy)$$

$$\frac{-2 + 2xy}{-M} = \frac{2}{y} = \frac{2}{y}$$

$$\mu(y) = e^{\int \frac{2}{y} dy} = \frac{1}{y^2}$$

② 乘 $\mu(y)$

$$(-\frac{1}{y} + x) dx + \frac{x}{y^2} dy = 0.$$

$$\text{即 } x dx + \frac{xdy - ydx}{y^2} = 0.$$

$$\frac{x^2}{2} - \frac{x}{y} = C$$

§5. 高阶: 降阶

$$\therefore \frac{d^2 y}{dx^2} = f(x)$$

$$\frac{dy}{dx} = \int f(x) dx + C_1$$

$$y = \int (\int f(x) dx + C_1) dx + C_2$$

$$\therefore \frac{d^2 y}{dx^2} = f(x, \underbrace{\frac{dy}{dx}}_p)$$

$$\begin{cases} \frac{d^2 y}{dx^2} = f(x, y, \frac{dy}{dx}) \\ y|_{x=x_0} = y_0 \\ \frac{dy}{dx}|_{x=x_0} = y_0' \end{cases}$$

$$\frac{d^2y}{dx^2} = \frac{dp}{dx}$$

$$\frac{dp}{dx} = f(x, p) \Rightarrow p = \varphi(x, c_1)$$

$$\frac{dy}{dx} = \varphi(x, c_1)$$

$$\Rightarrow y = \int \varphi(x, c_1) dx + c_2$$

$$y^{(n)} = f(x, y^{(n-1)}) \Rightarrow y^{(n)} = \frac{dp}{dx}.$$

n=2, 3, 4, 5.

$$\equiv \frac{d^2y}{dx^2} = f(y, \frac{dy}{dx})$$

$$\frac{dy}{dx} \cdot \frac{dp}{dy}$$

$$\frac{dy}{dx} = p$$

$$\frac{dp}{dx} = f(y, p)$$

$$\frac{dp}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{dp}{dx} = \frac{dp}{dy}$$

$$\frac{dy}{dx} = p \frac{dp}{dy}$$

$$p \frac{dp}{dy} = f(y, p)$$

$$\text{eg. } \left(\frac{dy}{dx}\right)^2 - y \frac{d^2y}{dx^2} = 0$$

$$\text{Then } p = \frac{dy}{dx}$$